

Econometrics Methods for Empirical Economics

Victoria Godio & Baptiste Messant

Toulouse School of Economics

January 2026

Contents

1 Exercise 1 : Estimation of a censored model **1**

1.1 Data preparation and descriptive analysis . . . 2

1.2 Baseline models . . . 4

1.3 Heckman selection model . . . 5

1.4 Identification robustness . . . 9

1.4.1 Model 1 with Nonlinearities . . . 12

1.4.2 Model 2 with Interactions . . . 13

1.4.3 Model 3 with Age Controls . . . 13

2 Exercise 2 : Backtesting Value-at-Risk models **20**

2.1 Question 1 . . . 20

2.2 Question 2 . . . 21

2.2.1 a) Historical quantile . . . 21

2.2.2 b) Gaussian quantile . . . 21

2.2.3 c) Standardized Student . . . 21

2.2.4 d) Extreme Value Approach . . . 22

2.2.5 e) Historical quantile on the returns . . . 23

2.2.6 VaR Computations . . . 24

2.3 Question 3 . . . 24

2.3.1 VaR Results . . . 24

2.4 Question 4 . . . 25

2.4.1 VaR plot . . . 25

2.4.2 Coverage Test . . . 28

3 Exercice 3 : GMM estimation of a demand and supply model **31**

3.1 Question 1,2,3 : Data Description . . . 31

3.2 Question 4 . . . 32

3.2.1 Additional Variables . . . 32

3.2.2 Instruments . . . 33

3.2.3 Results . . . 34

3.3 Question 5 . . . 37

3.3.1 Results . . . 38

3.4 Question 6 . . . 41

3.4.1 Results . . . 41

3.4.2 Demand coefficients . . . 42

3.4.3 Supply coefficients . . . 43

1 Exercise 1 : Estimation of a censored model

In this exercise, you estimate a Generalized Tobit suing the famous data about labor-force participation and wages of married women from Mroz (1987). The dataset is in the model section as well as the paper of Mroz. Here are the meaning of the different variables.

Motivation and interest of the question :

The empirical literature on married women's labor supply has long been marked by striking discrepancies in estimated income and substitution effects. These conflicting results raise a key question: Are these differences driven by real behavioral variations, or do they stem from methodological and statistical assumptions?

Mroz (1987)'s paper adresses this question by examining how economic and statistical assumptions influence the estimation of substitution and income effects in married women's labor supply models. The paper provides a systematic analysis of key assumptions commonly made in the existing literature by estimating a wide range of specifications on a single dataset (PSID 1975 labor supply data). This approach allows the author to isolate the role of econometric assumptions in explaining the large dispersion of results found in previous studies. In particular, the analysis focuses on three main issues: the exogeneity of wages and other covariates, the treatment of self-selection into the labor force, and the impact of controlling for taxes.

The Mroz's paper and the following exercise rely on data from the University of Michigan's Panel Study of Income Dynamics (PSID) from 1975, with interviews conducted in 1976. It was only in 1976 that the PSID began interviewing the wives of households, as well as the heads of households, unlike previous years. The data comprises 753 married white women aged 30–60.

Below is a summary of the variables used in the analysis.

Variable Summary

Table 1: Description of variables

Variable	Description
`lfp`	Dummy variable equal to 1 if the woman worked in 1975, 0 otherwise
`hours`	Wife's hours of work in 1975
`kidsl6`	Number of children under 6 years old in the household
`kids618`	Number of children aged 6–18 in the household
`age`	Wife's age
`educ`	Wife's educational attainment (years of schooling)
`wage`	Wife's average hourly earnings in 1975 (1975 dollars)
`wage76`	Wife's wage reported at the 1976 interview (for 1976)
`hhours`	Husband's hours worked in 1975

Variable	Description
`hage`	Husband's age
`heduc`	Husband's educational attainment (years of schooling)
`hwage`	Husband's wage in 1975 dollars
`faminc`	Family income in 1975 dollars
`mtr`	Marginal tax rate faced by the wife, including Social Security taxes
`mothereduc`	Wife's mother's years of education
`fathereduc`	Wife's father's years of education
`unemployment`	Unemployment rate in the county of residence
`bigcity`	Dummy variable equal to 1 if living in a large city (SMSA), 0 otherwise
`exper`	Actual years of the wife's previous labor market experience

Econometric Model

Let the latent selection index be defined as

$$d_i^* = z_i^\top \gamma, \qquad\qquad d_i = \mathbf{1}\{d_i^* > 0\}. \tag{1}$$

The wage equation (observed only when $d_i = 1$) is

$$y_i \equiv \ln(\text{wage}_i) = x_i^\top \beta + \varepsilon_i, \qquad \text{observed if } d_i = 1. \tag{2}$$

Assume that (u_i, ε_i) are jointly normally distributed with

$$\mathbb{E}[u_i] = \mathbb{E}[\varepsilon_i] = 0, \qquad \text{Var}(u_i) = 1, \qquad \text{Var}(\varepsilon_i) = \sigma^2, \qquad \text{Corr}(u_i, \varepsilon_i) = \rho. \tag{3}$$

In the first step, we choose the explanatory variables as

$$z = \Big(\text{kidsl6, kids618, log(hwage)}\Big), \tag{4}$$

$$x = \Big(\text{exper, educ, bigcity}\Big). \tag{5}$$

1.1 Data preparation and descriptive analysis

(a) *Report the counts by $d_i = lfp_i$ and summary statistics for wages among workers. Compare with Table III of the paper.*

For this question, we first count the number of women who worked in 1975 ($lfp = 1$) and the number of women who did not work ($lfp = 0$). We then report summary statistics for wages among working women.

Table 2: Descriptive statistics of the variable wage for working women

Statistic	**Value**
Number of working women ($lfp = 1$)	428.00
Mean wage	4.18
Standard deviation of wage	3.31
Minimum wage	0.13
1st quartile	2.26
Median wage	3.48
3rd quartile	4.97
Maximum wage	25.00

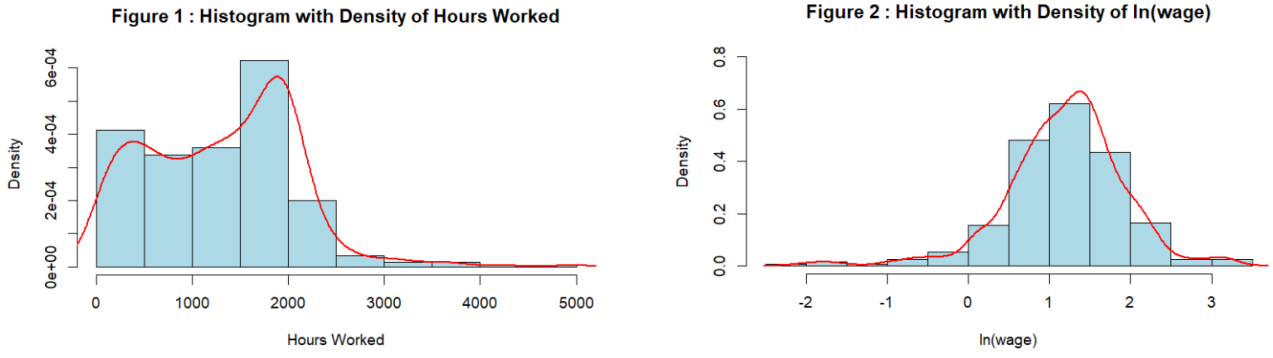
```
Labor force participation in the sample:

Out of 753 married women in the sample, 428 participate in the labor force
(lfp = 1), representing 56.839 % of the sample.
325 women do not participate in the labor force (lfp = 0), which corresponds
to 43.161 % of the sample.
```

Figure 1: Output of the Labor force participation from the sample

Our data is composed of an overall of 753 married white women between the ages of 30 and 60 in 1975. From it, we get 428 working women and 325 women that did not work in 1975, which is line with the results of the paper. In other words, in the data there is a proportion of 56.84% of working women and 43.16% of non-working women. Continuing with the descriptive statistics, we report the mean and standard deviation of the wage variable, defined as the average hourly earnings for women with a job. The average hourly wage is 4.18, with a standard deviation of 3.31, indicating a dispersion in earnings among married employed women in 1975. Additionally, we can remark that there is a big difference of wages between women with a minimum wage of 0.13 and a maximum wage of 25. The median is 3.48 and is lower than the mean suggesting that there is a small number of women with high earnings.

(b) *Plot the distribution of hours and of* ln(*wage*) *for workers.*



(a) Distribution of hours worked

(b) Distribution of ln(wage)

Figure 2: Distributions of hours worked and of ln(wage) for working women

Figure 1 shows the distribution of wives' annual hours of work in 1975. Most observations are concentrated below 2,000 hours per year, while relatively fewer women work more than 2,000 hours. The density curve displays a clear peak around 2,000 hours, indicating that this is the most common level of annual labor supply among working women.

Figure 2 shows the distribution of the natural logarithm of wives' average hourly wages in 1975 (in 1975 dollars). Most observations are concentrated between 1 and 2, indicating that the majority of women earn wages within this log range.

(c) *Discuss plausible exclusion restrictions for identification. Justify briefly from economic reasoning.*

As emphasized in Mroz (1987), wages are observed only for a subsample of women who participate in the labor force. Estimating wage by using this subsample does not control for self-selection into employment. In our case exclusion restriction can happen, it is caracterized when there is a variable that affect the probability of entering into the labor force, but should not have an impact on the wage. These variables serve as instruments to adress the selection bias. A plausible exclusion restriction can be the variable kidsl6, relating to the number of children under 6 years old in the household. Indeed, the presence of young children can have an impact on the women's likelihood of participing in the labor market. Young children require more supervision, making more harder for mothers to work outside the home. While kidsl6 affects the decision to work, it does not directly influence a woman's hourly wage if she is employed. Wages are primarily determined by her education, experience, and skills, not the number of young children.

1.2 Baseline models

(a) *Estimate OLS of* $\ln(wage)$ *on* x_i *using the working subsample. Interpret coefficients.*

Table 3: OLS Regression Results of ln(wage) on x_i

Term	Estimate	Std. Error	Statistic	p-value
Intercept	-0.4123	0.1909	-2.1596	0.0314
exper	0.0156	0.0040	3.8797	0.0001
educ	0.1075	0.0144	7.4873	0.0000
bigcity	0.0603	0.0683	0.8834	0.3775

According to the results obtained, we can interpret the coefficients obtained from the OLS regression, here the coefficients are a semi-elasticity of wage compared to the explanatory variables from the vector x . Among the three coeffcients tested only, bigcity is not significant at 1%. Hence, a variation of 1 unit of exper leads on average, ceteris paribus, to a variation of 1.56% of wage. A variation of 1 unit of educ leads on average, ceteris paribus, to a variation of 10.75% of wage.

(b) *Estimate a Probit model for participation d_i on z_i. Report average partial effects (APEs) and propose some measures of fit.*

Table 4: Probit Regression Results for Labor Force Participation

Term	Estimate	Std. Error	Statistic	p-value
Intercept	0.3332	0.1648	2.0224	0.0431
kidsl6	-0.5422	0.0949	-5.7114	0.0000
kids618	0.0161	0.0356	0.4521	0.6512
log_hwage	-0.0296	0.0800	-0.3698	0.7115

Table 5: Average Partial Effects (APE) for the Labor Force Participation

Term	APE
kidsl6	-0.2052
kids618	0.0061
log_hwage	-0.0112

For this question, we have conducted a Probit model and we computed the Average Partial Effects (APEs) to be able to interpret the magnitude of the coefficients of a Probit model. In short, among the three variables tested, we have only one that is significant kidsl6 at 1% confidence level, relating to the number of children under 6 years old in the household. To interpret the coefficient of this variable, we look at the APEs and therefore we can say that on average, an additional child under the age of 6 reduces the probability of working for women in 1975 by approximately 20.5 percentage points.

Additionnaly, we added some measures of fit : the Pseudo- R2 Mc Fadden equals to 0.034 and the model correctly predicts about 62.3% of labor force participation outcomes.

(c) *Explain why OLS on workers may be biased relative to the full-population wage equation.*

Estimating the wage equation using OLS only on working women may lead to biased results compared to the full population wage equation. This is because wages are observed only for women who participate in the labor force, while non-employed women are excluded from the sample. If women who work differ systematically from those who do not due to unobserved characteristics such as ability, motivation, or preferences then the subsample of employed women is not random. As shown by Heckman (1980) and emphasized in Mroz (1987), failing to control for self-selection into the labor force implies sample selection bias.

1.3 Heckman selection model

(a) *Estimate the two-step model. Report the Inverse Mills ratio and standard errors.*

Table 6: Probit Labor Participation Selection Model Results

Variable	**Estimate**	**Std. Error**	**Statistic**	**p-value**
Intercept	0.3332	0.1648	2.0224	0.0431
kidsl6	-0.5422	0.0949	-5.7114	0.0000
kids618	0.0161	0.0356	0.4521	0.6512
log_hwage	-0.0296	0.0800	-0.3698	0.7115

Table 7: OLS estimation of ln(wage) with selection correction

Term	**Estimate**	**Std. Error**	**Statistic**	**p-value**
(Intercept)	-0.3355	0.2370	-1.4158	0.1576
exper	0.0153	0.0041	3.7472	0.0002
educ	0.1087	0.0145	7.4784	0.0000
bigcity	0.0611	0.0683	0.8945	0.3716
IMR	-0.1322	0.2413	-0.5477	0.5842

For this question we have used the two steps model : composed of a selection equation (participation into the labor force for women in 1975), it estimates the probability of an observation being included in the sample and the outcome equation, who adjusts the main OLS regression using a correction term called Inverse Mills Ratio (IMR). IMR accounts for the non-random selection process in the analysis.

The first step consists of estimating the probit model for the participation of labor supply, then we computed the Inverse Mills ratio and finally estimate the OLS regression adjusted with the Inverse Mills Ratio that corrected for the selection bias. We have obtained from our code the estimators of the selection equation and the outcome equation composed of the variables exper, educ, bigcity and Inverse Mills Ratio (IMR). We can note that IMR is not significant (high p-value), indicating that we find no evidence of sample selection bias in the outcome equation.

(b) *Estimate the full-information MLE version (see the likelihood in any textbook or in the slides). Report the estimators of* ρ *and* σ*, and a test of* $H_0 : \rho = 0$*. Report both the Wald test and the LR test.*

For this question, the following code will estimate the Heckman selection model by using MLE. It jointly estimates a Probit model for labor force participation and the wage equation observed only for working women. The correlation between the unobserved determinants of participation and outcome equations is explicitly modeled. Standard errors are obtained from the numerical Hessian, and both Wald and Likelihood ratio tests are used to assess whether sample selection bias is present.

Table 8: MLE estimates for the participation equation

Variable	**Coef**	**SE**	z**-stat**	p**-value**
Intercept	0.1457	0.1519	0.9597	0.3372
kidsl6	-0.4060	0.0846	-4.7990	0.0000
kids618	0.0187	0.0305	0.6153	0.5383
log_hwage	0.0481	0.0714	0.6732	0.5008

Table 9: MLE estimates for the wage equation

Variable	**Coef**	**SE**	z**-stat**	p**-value**
Intercept	0.0167	0.1923	0.0869	0.9308
exper	0.0114	0.0041	2.7794	0.0054
educ	0.1128	0.0137	8.2586	0.0000
bigcity	0.0682	0.0664	1.0270	0.3044

Table 10: MLE estimates of σ and ρ

Parameter	**Estimate**	**SE**	z**-stat**	p**-value**
σ	0.8320	0.0452	18.3934	0
ρ	-0.8058	0.0485	-16.6218	0

Table 11: Tests of the null hypothesis $H_0 : \rho = 0$

Test	**Result**
Wald Test	
Estimated ρ	-0.8058
Standard error	0.0485
χ^2 statistic	276.2832
p-value	0.0000
Conclusion	Reject $H_0 \rightarrow$ significant selection bias
Likelihood Ratio (LR) Test	
Unrestricted log-likelihood	-924.5355
Restricted log-likelihood ($\rho = 0$)	-930.5326
LR χ^2 statistic	11.9941
p-value	0.0005
Conclusion	Reject $H_0 \rightarrow$ significant selection bias

According to the results obtained, we have found that with MLE estimation firstly only the variable kidsl6 is significant and reduces probability of working (very significant effect).

Secondly, for the second equation we have found that the variables exper (1% level) and educ (1% level) are both significant and has a positive impact on the wages.

Additionally, the LR and Wald tests both found that there is a significant selection bias in the data.

(c) *Compare the three different estimates of key returns (e.g. to education and experience). Comment on magnitudes and standard errors.*

In this question we have to compare the results obtained of the outcome equation. Through the exercise we have estimated this equation with three different methods : the first one with the naive estimation directly with OLS model, then the Heckman selection model taking into account the possible selection bias with the Inverse Mills Ratio and finally the Heckman selection model but estimated by MLE. To have a clearer view we have added a summary tables that summarizes the results of the outcome equation with the three different estimations.

Table 12: Comparison of key returns across estimation methods

	OLS	**Heckman two-step**	**Heckman MLE**
Experience	0.0156***	0.0153***	0.0114***
	(0.004)	(0.0041)	(0.0041)
Education	0.1075***	0.1087***	0.1128***
	(0.0144)	(0.0145)	(0.0137)
Big city	0.0603	0.0611	0.0682
	(0.0683)	(0.0683)	(0.0664)

Notes: Standard errors in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

From the estimation results, we can first note that for all three estimations : bigcity variable is not significant and exper and educ are significant at 1% level.

The variable exper is highly significant and positive in all specifications, but its magnitude decreases when correcting for sample selection, suggesting that OLS slightly overestimates the effect.

For the variable educ, the effect of an additional year of the wife's educational attainment is positive and statistically significant across all estimation methods. In contrast to the results for experience, the magnitude of the coefficient is highest in the Heckman MLE estimation and lowest in the OLS specification. This suggests that correcting for sample selection slightly increases the estimated return to education.

For the standard errors, we can note that for the significant variables : educ and exper, the standard errors do not change a lot and remain very similar across the three models.

(d) *Compute and interpret predicted participation probabilities and expected wages for workers for two contrasting profiles (define and report them).*

For this question, we compute predicted participation probabilities and expected wages for two contrasting profiles of women. We therefore construct two very different types of women. Profile A corresponds to women with characteristics that increase the probability of labor force participation. Specifically, we set the number of young children to zero and the number of

children aged 6 to 18 to its 25th percentile. We assign a low husband's wage (25th percentile of log(hwage)), a high level of labor market experience (75th percentile), 15 years of education, and assume residence in a big city (bigcity = 1).

Profile B, in contrast, corresponds to women with characteristics that decrease the probability of participation. We assign two young children, the 75th percentile of children aged 6 to 18, a high husband's wage (75th percentile of log(hwage)), low labor market experience (25th percentile), 7 years of education, and assume that the woman does not live in a big city (bigcity = 0).

The R code for this question constructs the two representative profiles of women and uses the estimated Heckman model to predict their labor force participation and their expected wages. Using the MLE estimates of the selection equation, it computes for each profile the probability of working and the IMR. These are then used to get the expected wages from either the Heckman two-steps or from the Heckman MLE model.

Table 13: Summary Table of the Two Profiles' Characteristics

Profile	**kidsl6**	**kids618**	**log_hwage**	**exper**	**educ**	**bigcity**
Profile A (high participation)	0	0	1.566175	15	15	1
Profile B (low participation)	2	2	2.215577	4	7	0

Table 14: Participation Probabilities and Expected Conditional Wages

Profile	Pr_working	IMR	Salaire_2step	Salaire_MLE
Profil A (high partipation profile)	0.5875	0.6627	4.4680	4.4998
Profil B (low participation profile)	0.3007	1.1574	1.3954	1.0792

Hence, Profile A (favorable to work) has a predicted probability of labor force participation of about 59%, while Profile B (with less favorable characteristics) has a participation probability of about 30%. Conditional on working, the two-step Heckman model predicts an expected hourly wage of 4.47 for Profile A, whereas Profile B has an expected wage of 1.39. Using the MLE specification, Profile A is expected to earn an hourly wage of about 4.50 (very close to the two-step estimate), while Profile B is expected to earn about 1.08. Overall, Profile A is both more likely to work and, conditional on working, expected to earn a substantially higher wage than Profile B under both estimation methods.

1.4 Identification robustness

(a) *Update the values of x and z in order to improve the model*

The goal of this question is to improve the model by adding new variables in x and z. Indeed, since the beginning of the exercise we determine the participation into the labor force for women in 1975 only with three variables: kidsl6, kids618 and log(hwage), and we have

estimated the log(wage) also with only three variables: exper, educ and bigcity. Adding new variables for both equations allows for a better identification.

Firstly, to estimate the probability to participate into the labor force for the vector z, we can add as variables: unemployment, fathereduc and mothereduc. Indeed, the unemployment rate could affect the likelihood of finding a job, and therefore influence the decision to participate in the labor market. A high unemployment rate reflects a tight labor market with greater competition for jobs, which may discourage labor force participation. Conditional on employment, however, local labor market conditions are not expected to directly determine an individual's hourly wage, which is primarily driven by personal characteristics such as education and experience.

We also wanted to add the father's education level and the mother's education level. Indeed, the level of education of the parents can influence the willingness to participate or not in the labor force, for example through family values or attitudes toward work. Women with highly educated parents may be more encouraged to work. However, we do not expect that the education of the parents can directly influence wages.

For the wage equation, we decided to include the age of the woman. Age can have a direct impact on the level of wages, as older individuals typically have a higher labor productivity, leading to higher wages.

In short, the new vectors x and z are the following:

Selection equation (labor force participation) The vector of explanatory variables for the selection equation is:

$$z = (\text{kidsl6}, \text{kids618}, \log(\text{hwage}), \text{unemployment}, \text{fathereduc}, \text{mothereduc})$$

Outcome equation (wage equation) The vector of explanatory variables for the wage equation is:

$$x = (\text{exper}, \text{educ}, \text{bigcity}, \text{age})$$

(b) *Report the same quantities than above, comment*

For this question, we reuse the code from Question 3(b) to estimate log(wage), but with the new vectors x and z, using a Heckman selection model estimated by MLE.

Table 15: Labor Force Participation Results with the Extended Model

Variable	Coef	SE	z_stat	p_value
Intercept	0.0351	0.2242	0.1564	0.8757
kidsl6	-0.5788	0.0969	-5.9717	0.0000
kids618	0.0161	0.0356	0.4520	0.6513
log_hwage	-0.0681	0.0851	-0.7997	0.4239
Unemployment	-0.0118	0.0153	-0.7738	0.4390
fathereduc	0.0117	0.0164	0.7139	0.4753
mothereduc	0.0410	0.0172	2.3892	0.0169

Table 16: Wage Equation Results with the Extended Model

Variable	Coef	SE	z_stat	p_value
Intercept	-0.3924	0.2926	-1.3411	0.1799
exper	0.0165	0.0046	3.6052	0.0003
educ	0.1074	0.0144	7.4788	0.0000
bigcity	0.0625	0.0685	0.9124	0.3616
age	-0.0017	0.0048	-0.3584	0.7201

Table 17: Parameter Estimates with the Extended Model

Parameter	Estimate	SE	z_stat	p_value
σ	0.6675	0.0242	27.5577	0.0000
ρ	0.0926	0.2298	0.4031	0.6869

Table 18: Tests for $H_0: \rho = 0$

Wald Test	
Estimated ρ	0.0926
χ^2 statistic	0.1625
P-value	0.6869
Conclusion	Failure to reject $H_0 \rightarrow$ no significant selection bias
Likelihood Ratio (LR) Test	
Unrestricted log-likelihood	-924.4560
Restricted log-likelihood ($\rho = 0$)	-924.5242
χ^2 LR statistic	0.1365
P-value	0.7118
Conclusion	Failure to reject $H_0 \rightarrow$ no significant selection bias

For the labor force participation equation, only the variables kidsl6 and mothereduc are statistically significant at the 1% and 5% levels, respectively. The coefficient for kidsl6 remains

negative, consistent with the results from Question 3.b. The coefficient for mothereduc is positive, suggesting that an additional year of the wife's mother's education increases the expected labor force participation of the wife.

For the wage equation, the added variable age is not statistically significant in explaining the expected wage levels of married women.

Additionally, the Wald and Likelihood Ratio tests fail to reject the null hypothesis of no selection bias, in contrast to the results obtained in Question 3(b) with the simpler model. This suggests that, once additional variables are included, selection into employment is no longer statistically significant, indicating an improved model and better identification.

(c) *Functional-form robustness: include now nonlinearities (e.g., exper^2), interactions, or alternative age controls. Summarize differences.*

In this question we will continue to modify the vector x and z by adding interactions, non linearities and alternative age controls. We can try several models trying to improve the model and compare them with the models obtained from Questions 3(b) and 4(b). For this question we will test three different models: Model 1 with non linearities, Model 2 with interactions, and Model 3 with age controls.

1.4.1 Model 1 with Nonlinearities

For Model 1 therefore, we have:

We keep the same vector of explanatory variables as in the baseline specification:

$$z = (\text{kidsl6},\ \text{kids618},\ \log(\text{hwage}),\ \text{unemployment},\ \text{fathereduc},\ \text{mothereduc}).$$

To allow for non-linear relationships, we added to the wage equation quadratic terms in experience and age:

$$x = (\text{exper},\ \text{exper}^2,\ \text{educ},\ \text{bigcity},\ \text{age},\ \text{age}^2).$$

We decided to test this new model. Indeed, Model 1 allows us to relax the assumption that wages are linear over the life cycle. We assume here that the level of experience and the age of workers are not necessarily linear with wages. Returns to the labor market are expected to be diminishing: workers accumulate skills rapidly at the beginning of their careers, while additional experience yields smaller productivity gains at later stages. Including a squared term in experience captures this non-linear justification.

Similarly, the effect of age on wages is unlikely to be linear. Age captures not only experience, but also seniority and career progression, which may lead to wage growth at early and mid-career stages and a stabilization later on.

1.4.2 Model 2 with Interactions

For Model 2, we decided to introduce interactions between specific variables, and we obtain the new vectors x and z.

We keep the same vector of explanatory variables as in the baseline specification:

$$z = (\text{kidsl6},\ \text{kids618},\ \log(\text{hwage}),\ \text{unemployment},\ \text{fathereduc},\ \text{mothereduc}).$$

And we change the vector x:

$$x = (\text{exper},\ \text{educ},\ \text{bigcity},\ \text{age},\ \text{educ} \times \text{exper},\ \text{bigcity} \times \text{educ}).$$

Here we introduced two new terms: educ × exper and bigcity × educ. First, the interaction between education and experience captures the idea that the return to experience may depend on the level of education. More educated individuals are more likely to work in occupations with steeper career paths and greater opportunities for learning and promotion, so additional experience may translate into higher wage gains for them.

Additionally, the second interaction term between big city and education indicates the possibility that the return to education can differ depending on geographical labor markets. Indeed, large cities typically offer a wider range of demand for high-skilled jobs (especially in the service sector), implying an increase in wages associated with a high level of education.

1.4.3 Model 3 with Age Controls

We keep the same vector of explanatory variables as in the baseline specification:

$$z = (\text{kidsl6},\ \text{kids618},\ \log(\text{hwage}),\ \text{unemployment},\ \text{fathereduc},\ \text{mothereduc}).$$

We change the composition of the vector x by replacing the continuous age variable with age-group dummies into three categories, with women aged between 30–39 years as the reference group:

$$x = (\text{exper},\ \text{educ},\ \text{bigcity},\ \text{age_40–49},\ \text{age_50–more}).$$

We choose this model and decide to replace the continuous age variable with age-group dummies. Using age groups allows us to study the potential non-linear relationship between age and wages and is particularly useful if wage growth occurs in discrete steps.

We decided to split the continuous age variable into three groups:

- 30 to 39 years to capture early career progression,
- 40 to 49 years to capture mid-career stabilization,
- 50 and more to capture later career phases.

We define the group of women aged 30–39 as the reference group.

After defining all three new models, the following code estimates each model using the Heckman MLE method and compares them with the models obtained in Questions 3(b) and 4(d). We reuse the same code as in the previous questions and adapt it accordingly.

- **Model 1 with Nonlinearities MLE results**

Table 19: Labor Force Participation Results with Model 1

Variable	**Coef**	**SE**	**z_stat**	**p_value**
Intercept	0.0386	0.2242	0.1721	0.8634
kidsl6	-0.5782	0.0967	-5.9816	0.0000
kids618	0.0163	0.0356	0.4575	0.6473
log_hwage	-0.0720	0.0844	-0.8537	0.3933
Unemployment	-0.0115	0.0153	-0.7515	0.4524
fathereduc	0.0117	0.0164	0.7128	0.4760
mothereduc	0.0412	0.0171	2.4033	0.0163

Table 20: Wage Equation Results with Model 1

Variable	**Coef**	**SE**	**z_stat**	**p_value**
Intercept	-1.4271	1.0224	-1.3959	0.1628
exper	0.0384	0.0135	2.8359	0.0046
exper2	-0.0007	0.0004	-1.6672	0.0955
educ	0.1066	0.0143	7.4544	0.0000
bigcity	0.0524	0.0683	0.7677	0.4427
age	0.0409	0.0478	0.8563	0.3918
age2	-0.0005	0.0006	-0.8600	0.3898

Table 21: Parameter Estimates with Model 1

Parameter	**Estimate**	**SE**	**z_stat**	**p_value**
σ	0.6652	0.0251	26.5228	0.0000
ρ	0.1273	0.2166	0.5879	0.5566

Table 22: Tests for $H_0: \ \rho = 0$

Wald Test	
Estimated ρ	0.1273
χ^2 statistic	0.3456
P-value	0.5566
Conclusion	Failure to reject $H_0 \rightarrow$ no significant selection bias
Likelihood Ratio (LR) Test	
Unrestricted log-likelihood	-922.0648
Restricted log-likelihood ($\rho = 0$)	-922.2105
χ^2 LR statistic	0.2913
P-value	0.5894
Conclusion	Failure to reject $H_0 \rightarrow$ no significant selection bias

From the result tables, we see that for the outcome equation, exper and educ are significant (1%) level and exper2 is significant (10%). Additionally, the coefficient of exper2 is negative implying that there are decreasing return from experience. The marginal effect is positive on the wage but it diminishes over time. Both the Wald test and the likelihood ratio test fail to reject the null hypothesis H: $= 0$. This suggests that there is no significant selection bias in the Model 1 specification.

- **Model 2 with Interactions MLE results**

Table 23: Labor Force Participation Results with Model 2

Variable	**Coef**	**SE**	**z_stat**	**p_value**
Intercept	0.0468	0.2242	0.2089	0.8345
kidsl6	-0.5787	0.0969	-5.9719	0.0000
kids618	0.0155	0.0356	0.4357	0.6631
log_hwage	-0.0707	0.0851	-0.8312	0.4058
Unemployment	-0.0120	0.0153	-0.7846	0.4327
fathereduc	0.0114	0.0164	0.6951	0.4870
mothereduc	0.0409	0.0172	2.3805	0.0173

Table 24: Wage Equation Estimates with Model 2

Variable	**Coef**	**SE**	**z_stat**	**p_value**
Intercept	-0.3426	0.4745	-0.7219	0.4704
exper	0.0093	0.0219	0.4225	0.6727
educ	0.1037	0.0332	3.1275	0.0018
bigcity	0.1361	0.3769	0.3611	0.7181
age	-0.0018	0.0048	-0.3747	0.7079
exper_educ	0.0006	0.0017	0.3389	0.7347
bigcity_educ	-0.0060	0.0298	-0.2025	0.8396

Table 25: Structural Parameter Estimates

Parameter	**Estimate**	**SE**	**z_stat**	**p_value**
σ	0.6674	0.0243	27.5156	0.00
ρ	0.0944	0.2288	0.4125	0.68

Table 26: Tests for $H_0: \ \rho = 0$

Wald Test	
Estimated ρ	0.0944
χ^2 statistic	0.1701
P-value	0.6800
Conclusion	Failure to reject $H_0 \rightarrow$ no significant selection bias
Likelihood Ratio (LR) Test	
Unrestricted log-likelihood	-924.3713
Restricted log-likelihood ($\rho = 0$)	-924.4446
χ^2 LR statistic	0.1465
P-value	0.7019
Conclusion	Failure to reject $H_0 \rightarrow$ no significant selection bias

We get the results from the model 2, we can directly remark that the two interaction variables are not significant at all, implying that the return of education do not depend on the level of experience, and the return of education on the wages do not depend also of living in a big city. Educ is the only variable that is significant in this model. This could be explained that once education is included in the wage equation, allowing its effect to vary with experience or location does not provide additional explanatory power.

Additionally, as seen in model 1 the Wald and LR tests refute to reject the null hypothesis, implying that there is no significant selection bias in this model.

- **Model 3 with Age Controls MLE results**

Table 27: Labor Force Participation Equation Results with Model 3

Variable	**Coef**	**SE**	**z_stat**	**p_value**
Intercept	0.0481	0.2241	0.2148	0.8299
kidsl6	-0.5787	0.0966	-5.9884	0.0000
kids618	0.0163	0.0356	0.4565	0.6481
log_hwage	-0.0730	0.0845	-0.8649	0.3871
Unemployment	-0.0119	0.0153	-0.7779	0.4366
fathereduc	0.0114	0.0164	0.6958	0.4866
mothereduc	0.0410	0.0171	2.3949	0.0166

Table 28: Wage Equation Estimates with Model 3

Variable	**Coef**	**SE**	**z_stat**	**p_value**
Intercept	-0.4884	0.2202	-2.2180	0.0266
exper	0.0166	0.0045	3.7133	0.0002
educ	0.1073	0.0143	7.5025	0.0000
bigcity	0.0489	0.0686	0.7121	0.4764
mid_career	0.0780	0.0746	1.0467	0.2952
later_career	-0.0648	0.1013	-0.6395	0.5225

Table 29: Parameter Estimates with Model 3

Parameter	**Estimate**	**SE**	**z_stat**	**p_value**
σ	0.6668	0.0249	26.7282	0.0000
ρ	0.1216	0.2173	0.5594	0.5759

Table 30: Tests for $H_0 : \ \rho = 0$

Wald Test	
Estimated ρ	0.1216
χ^2 statistic	0.3129
P-value	0.5759
Conclusion	Failure to reject $H_0 \rightarrow$ no significant selection bias
Likelihood Ratio (LR) Test	
Unrestricted log-likelihood	-923.2129
Restricted log-likelihood ($\rho = 0$)	-923.3477
χ^2 LR statistic	0.2697
P-value	0.6036
Conclusion	Failure to reject $H_0 \rightarrow$ no significant selection bias

According to the results for model 3, the results show that experience and education have a positive and statistically significant effect on wages, confirming the importance of human capital accumulation. In contrast, age group dummies are not statistically significant, indicating that age does not affect wages. Indeed, this result could be explained that since experience is included in the wage equation, most of the wage variation associated with age is already captured by this variable. Also, living in a large city has a positive but insignificant effect on wages. Additionally, as in model 1 and 2, the Wald and LR test failed to reject the null implying that there is no significant selection bias.
Last step of the question is to compare these 3 models that we have created and compare to simpler models from the questions 3b and 4b.

Let's recall the results obtained from the model of question 3b) :

For the outcome equation, we have found that educ has a positive coefficient and is very significant (1%) and exper has also a positive coefficient and is significant (1%). And with this model, the Wald and LR tests refutes the null, implying that with this model we have a significant selection bias.

Let's recall the results obtained from the model of question 4b) : For the model done in question 4b), we have included variables in the selection equation : unemployment, fathereduc and mothereduc and for the outcome equation we included the age. For the outcome equation, we have found that the variables exper and educ are both positive and very significant (1%), however age in the equation is not significant. Moreover, with the Wald and LR tests there is no selection bias compared to the model in 3b). Additionally, the model 4b gives a better results then the model 3b the variables, exper and educ are more significant in the model 4b.

To compare also with the three models created in the question. The model 2 with inter-

actions does not bring new information (the interactions terms adeed are not significant), for model 3 we did not find significant effect of the age (when using dummies for ages group). Lastly the Model 1 added non lineairities for the age and exper, and we found that exper and educ are highly significant and exper2 is significant (10%). Additionally, the coefficient of exper2 is negative implying that there are decreasing return from experience. So the Model 1 gives us more information concerning the effect of the experience on the wages.

In short, Model 4a and Model 1 provide the most satisfactory performance among the other models considered, both models fail to reject the null hypothesis of no selection bias. Both have very significant results for the variables exper and educ and a benefit of Model 1 is to gain information concerning the nonlinear effect of experience on the wages.

2 Exercise 2 : Backtesting Value-at-Risk models

The goal of this exercise is to calculate different Value-at-Risk (VaR) models. I provide, in a file, four historical series of daily exchange rates together with an estimate of their volatility (reported as the square), computed the day before for the current observation and estimated from a GARCH model. Throughout this exercise, we assume that this volatility is the true volatility (i.e., we ignore the uncertainty generated by the estimation step). The innovation is also reported, that is,

$$\frac{r_t}{\sigma_t} = \varepsilon_t.$$

Let r_t denote the (log) return of a financial asset. For a given tail probability $\alpha \in (0,1)$ (we choose $\alpha = 3\%$ in this exercise) and horizon h (here $h = 1$), the Value at Risk at level α is the upper bound on losses that will not be exceeded with probability α:

$$\mathrm{VaR}_{\alpha,h} = \sup\left\{r \in \mathbb{R} \ : \ \mathbb{P}(r_{t+h} \leq r) \leq \alpha\right\}.$$

Equivalently, $\mathrm{VaR}_{\alpha,h}$ is the α-quantile of the return distribution over horizon h.

Motivation and interest of the question :

Value at Risk (VaR) is one of the most important risk indicators in the financial world. It summarises how large a loss could be over a given time period under 'normal' market conditions for a chosen confidence level. In finance, it is generally used to estimate potential losses over a short time horizon. However, Value at Risk is also key to:

- Summarise the risks of a portfolio to make risk reporting accessible to people without a good understanding of risk management, such as the management team or the CEO.
- Set a risk limit to restrict trader activity and achieve a favourable risk-benefit ratio.
- Distribute the risk across sectors and/or portfolios.

For all these reasons, it is crucial to have the best possible estimate of value at risk. Having inaccurate estimates can lead to poor risk management decisions, such as under- or over-budgeting, and a larger risk-benefit trade-off. To perform this analysis, we use the UK/US interest rate, together with its standard deviation and residuals.

2.1 Question 1

In this question we choose to use the US/UK exchange rate for the following computations and analysis.

2.2 Question 2

2.2.1 a) Historical quantile

To compute q_{hist}, we use the `quantile` function on e_t with the specification `probs = 0.03` in order to obtain the empirical 3% quantile of e_t.

Here, we found that the empirical 3% quantile is $q_{\text{hist}} = -2.087$.

2.2.2 b) Gaussian quantile

To compute q_{gauss}, we use the `qnorm` function on e_t with `0.03` to indicate that we want the 3% quantile. The options `mean = 0` and `sd = 1` specify that we use a standard normal distribution to compute the 3% quantile.

Here, we found that the Gaussian 3% quantile is $q_{\text{gauss}} = -1.880$.

2.2.3 c) Standardized Student

In this section we estimate the degrees of freedom ν of a **variance-standardized Student-t** model for the residuals `e_t`, and then uses the fitted model to compute the 3% **quantile**. It first defines `fd_grad()`, which approximates the gradient of a scalar objective function using **centered finite differences**:

$$\frac{\partial}{\partial \theta_j} f(\theta) \;\approx\; \frac{f(\theta + h e_j) - f(\theta - h e_j)}{2h}.$$

The core is `loglik_std_t(nu, e)`: it assumes the errors follow a Student-t distribution with ν degrees of freedom, rescaled to have **unit variance**. Since a raw Student-t variable $T \sim t_\nu$ has variance $\text{Var}(T) = \nu/(\nu-2)$ for $\nu > 2$, the code uses the scale

$$s = \sqrt{\frac{\nu - 2}{\nu}}$$

and defines the standardized error as $e = sT$, so that $\text{Var}(e) = 1$. Writing $u = e/s$, the change of variables implies the density transformation

$$\log f_e(e) = \log f_T(u) - \log(s),$$

which explains the Jacobian term $-\log(s)$ added to the usual Student-t log-density. The function `negll_student()` returns the **negative** total log-likelihood, i.e. $-\sum_i \log f_e(e_i)$, so minimizing it is equivalent to maximum likelihood. The optimizer `nloptr()` (LBFGS) then searches for the MLE $\hat{\nu}$ within bounds $[2.01, 200]$ (the lower bound enforces $\nu > 2$ so the

variance exists). Finally, using $\hat{s} = \sqrt{(\hat{\nu}-2)/\hat{\nu}}$, the standardized 3% quantile is computed as

$$q_{0.03} = \hat{s}\, t_{\hat{\nu}}^{-1}(0.03),$$

which is exactly what the code stores in `q_student`.

Here we found that the Standardized 3% quantile is -1.896.

2.2.4 d) Extreme Value Approach

In this part we compute the 3% **left-tail quantile** using an **Extreme Value Theory (EVT)** approach based on the **Peaks-Over-Threshold (POT)** method with a **Generalized Pareto Distribution (GPD)**. It sets the tail probability $\alpha = 0.03$ and chooses a positive threshold $T = 1.5$ applied to losses: since we are interested in the left tail of $\varepsilon_t =$ `bdd$e_t`, exceedances are defined by $\varepsilon_t < -T$. The indices of such tail events are stored in `idx`, with $t^{=\#\{t:\varepsilon_t<-T\}}$ and t_{tot} total observations, so the empirical exceedance rate is

$$k_T = \frac{t}{t_{\text{tot}}} \approx \mathbb{P}(\varepsilon < -T).$$

To fit a GPD, the code converts left-tail exceedances into positive threshold exceedances: it sets $z = -\varepsilon$ (so large z corresponds to extreme negative ε), then defines excesses over the threshold as

$$y = z - T \quad \text{for } z > T,$$

so $y \geq 0$ are the POT excesses. The function `gpd_nll_grad()` implements the **negative log-likelihood** of a GPD for $y_1, \ldots, y_n$ with scale $\beta > 0$ and shape ξ, using a reparameterization $\beta = \exp(\theta)$ to ensure positivity. For $\xi \neq 0$, the GPD log-likelihood (up to the sign) is based on the term

$$1 + \xi \frac{y_i}{\beta} > 0,$$

and the negative log-likelihood minimized by the optimizer is

$$\text{NLL}(\beta, \xi) = n \log \beta + \left(\frac{1}{\xi} + 1\right) \sum_{i=1}^{n} \log\left(1 + \xi \frac{y_i}{\beta}\right).$$

The code also includes the exponential-limit case when $|\xi|$ is very small: as $\xi \to 0$, the GPD becomes exponential and

$$\text{NLL}(\beta, 0) = n \log \beta + \frac{1}{\beta} \sum_{i=1}^{n} y_i.$$

Analytical gradients with respect to (θ, ξ) are provided to `nloptr()` (LBFGS) to speed up maximum likelihood estimation, producing $\hat{\beta}$ and $\hat{\xi}$. Finally, it converts the fitted GPD tail

model into an estimate of the α-quantile of the original residuals. Using the POT approximation

$$\mathbb{P}(Z > z) \approx k_T \left(1 + \xi \frac{z - T}{\beta}\right)^{-1/\xi} \qquad (z > T),$$

solving $\mathbb{P}(Z > z_\alpha) = \alpha$ gives the EVT quantile on the $z = -\varepsilon$ scale:

$$z_\alpha = \begin{cases} T + \beta \log\left(\frac{k_T}{\alpha}\right), & \xi = 0, \\ T + \frac{\beta}{\xi}\left[\left(\frac{k_T}{\alpha}\right)^{\xi} - 1\right], & \xi \neq 0, \end{cases}$$

and the corresponding **left-tail quantile** for ε is $q_{\text{EVT}} = -z_\alpha$, stored as `q_evt` and printed at the end.

Interpretation of $1/\hat{\xi}$ In the GPD (POT) model, ξ is the *shape* parameter and it determines the tail behaviour of exceedances. Here,

$$\hat{\xi} = -0.0986 \quad \Rightarrow \quad \frac{1}{\hat{\xi}} = -10.142.$$

Because $\hat{\xi} < 0$, the fitted tail is *bounded* (finite endpoint). In this case, $1/\hat{\xi}$ should not be interpreted as a Pareto tail index (that interpretation is relevant when $\xi > 0$). The key implication of a negative ξ is that exceedances have a finite upper bound given by the GPD support condition, namely

$$y_{\max} = -\frac{\hat{\beta}}{\hat{\xi}} \quad \text{with} \quad \hat{\beta} = 0.6287,$$

so the model implies a maximum possible exceedance and therefore a finite bound on extreme losses within the fitted EVT specification.

Here we found that the EVT 3% quantile is -2.07.

2.2.5 e) Historical quantile on the returns

Here we have computed the 3% quantile of `r_t` to obtain the **historical VaR on returns**. As before, we use the `quantile` function with the specification `probs = 0.03` to compute the 3% quantile.

Here we found that the historical quantile on returns is $q_{\text{hist},r} = -0.0136$.

2.2.6 VaR Computations

Using the formula provided at the beginning of the exercise:

$$\text{VaR}_{0.03,t} = \sigma_t \, q_{0.03}.$$

we compute the VaR for all five estimated quantiles found above.

In this way, we have created five new columns:

- `VaR_hist` using the **historical** 3% quantile of `e_t`
- `VaR_gauss` using the **Gaussian** 3% quantile of `e_t`
- `VaR_student` using the 3% quantile of `e_t` under a **variance-standardized Student-t** model
- `VaR_evt` using the approximated 3% quantile of `e_t` from the **Extreme Value Theory** approach
- `VaR_hist_r` using the **historical** 3% quantile of `r_t`

2.3 Question 3

2.3.1 VaR Results

In this table, we report all the VaR results along with the corresponding `r_t`, `e_t`, and σ_t over time.

Here we have an overview of the first 10 lines of the results table

Table 31: Results table

time	r_t	sigma_t	e_t	VaR_hist	VaR_gauss	VaR_student	VaR_evt	VaR_hist_r
1	-0.003555	5.1e-05	-0.498823	-0.014879	-0.013405	-0.013517	-0.014793	-0.013632
2	0.014254	4.8e-05	2.066027	-0.014403	-0.012976	-0.013084	-0.014320	-0.013632
3	-0.004439	6.1e-05	-0.567477	-0.016331	-0.014714	-0.014836	-0.016237	-0.013632
4	0.010256	5.8e-05	1.351413	-0.015844	-0.014274	-0.014393	-0.015752	-0.013632
5	0.016776	6.2e-05	2.135701	-0.016398	-0.014774	-0.014896	-0.016303	-0.013632
6	0.003690	8.1e-05	0.410766	-0.018754	-0.016896	-0.017036	-0.018645	-0.013632
7	-0.009410	7.4e-05	-1.090224	-0.018019	-0.016234	-0.016369	-0.017915	-0.013632
8	-0.010519	7.5e-05	-1.211383	-0.018127	-0.016332	-0.016467	-0.018023	-0.013632
9	-0.002957	7.8e-05	-0.334340	-0.018461	-0.016632	-0.016770	-0.018354	-0.013632
10	0.001076	7.2e-05	0.126912	-0.017702	-0.015948	-0.016081	-0.017599	-0.013632

To compare the different VaR estimates, we compute the mean of each series and report them in the results table. We observe that the **historical** and the **EVT** VaR have almost the same value. We also see that the **Gaussian** and the **Student** VaR are approximately equal. Finally, the **historical VaR on returns** lies between the other values.

Table 32: Mean VaR estimates

VaR	Mean
Hist	-0.014134
Gauss	-0.012733
Student	-0.012839
EVT	-0.014052
Hist_r	-0.013632

2.4 Question 4

2.4.1 VaR plot

Before starting Question 4, we thought it would be interesting to examine how often the VaR is violated under each model (i.e., the number of times r_t falls below the VaR). To do so, we plotted each `VaR` series together with `r_t`, and we added a red dot for each violation.

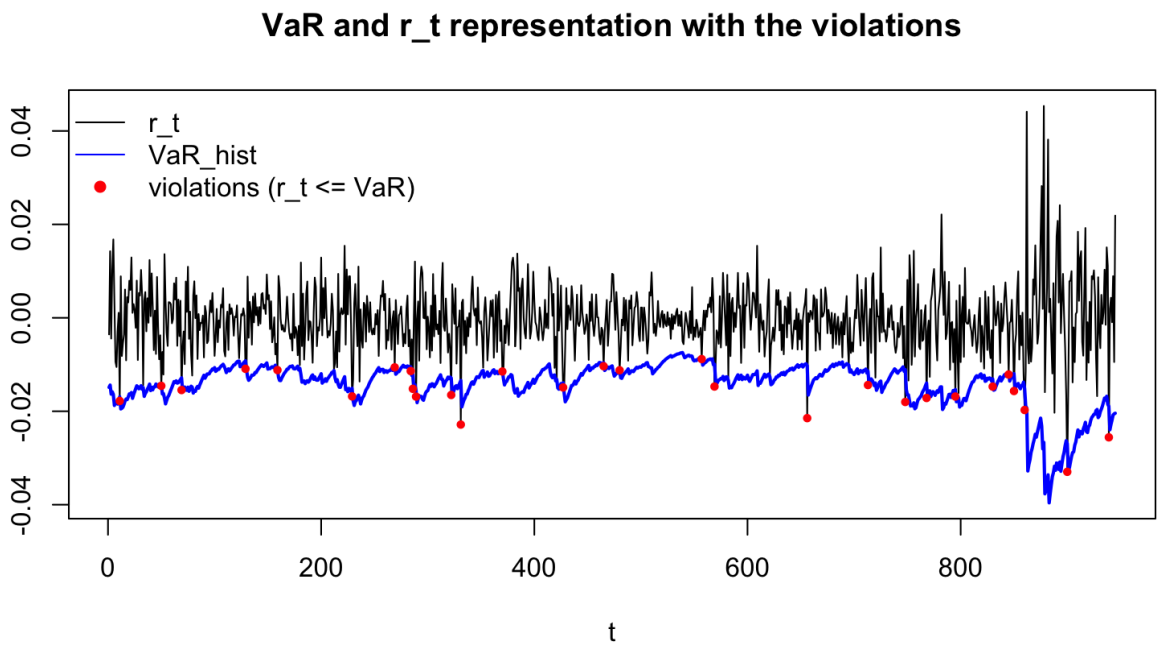


Figure 3: VaR_hist and r_t with violations (red dots).

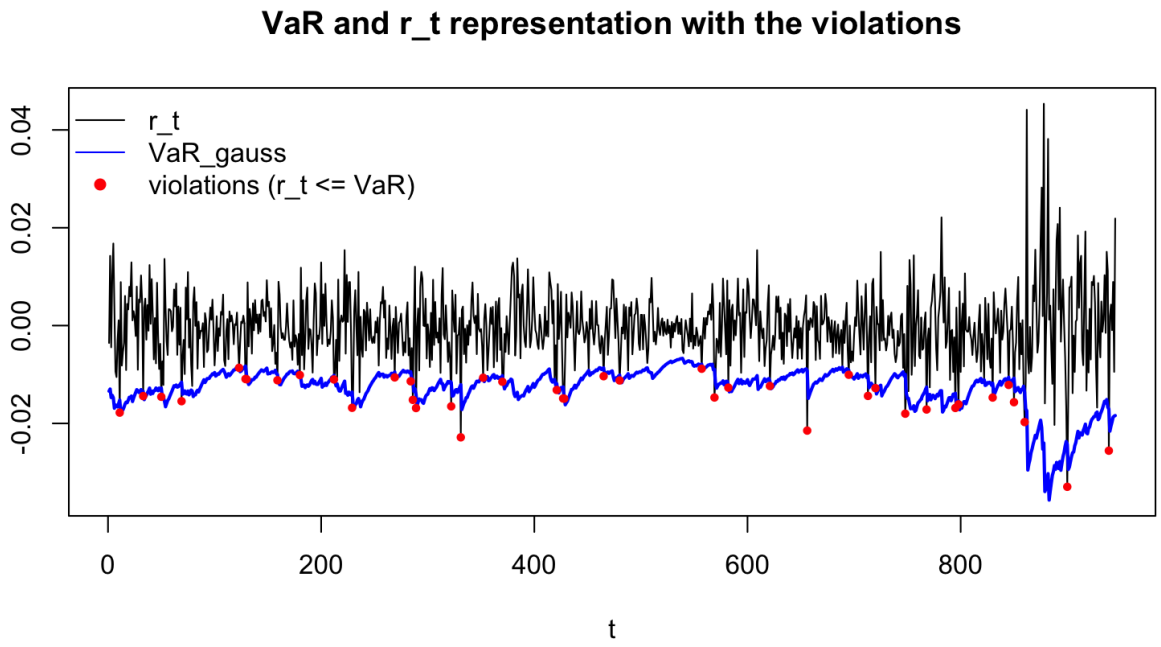


Figure 4: VaR_gauss and r_t with violations (red dots).

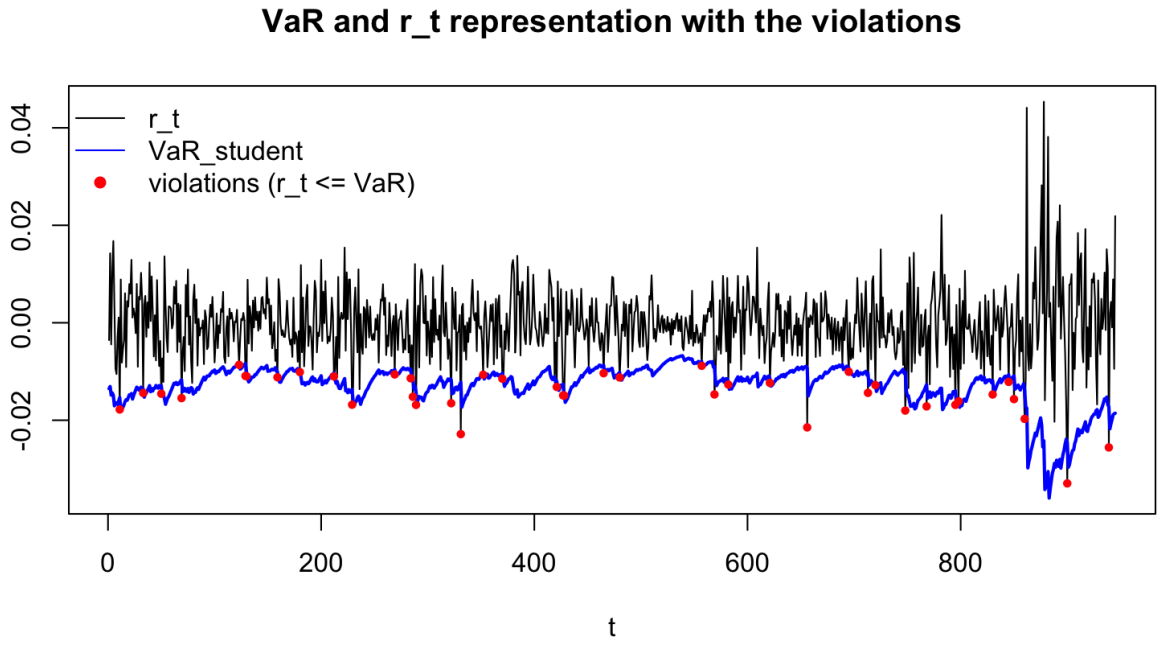


Figure 5: VaR_student and r_t with violations (red dots).

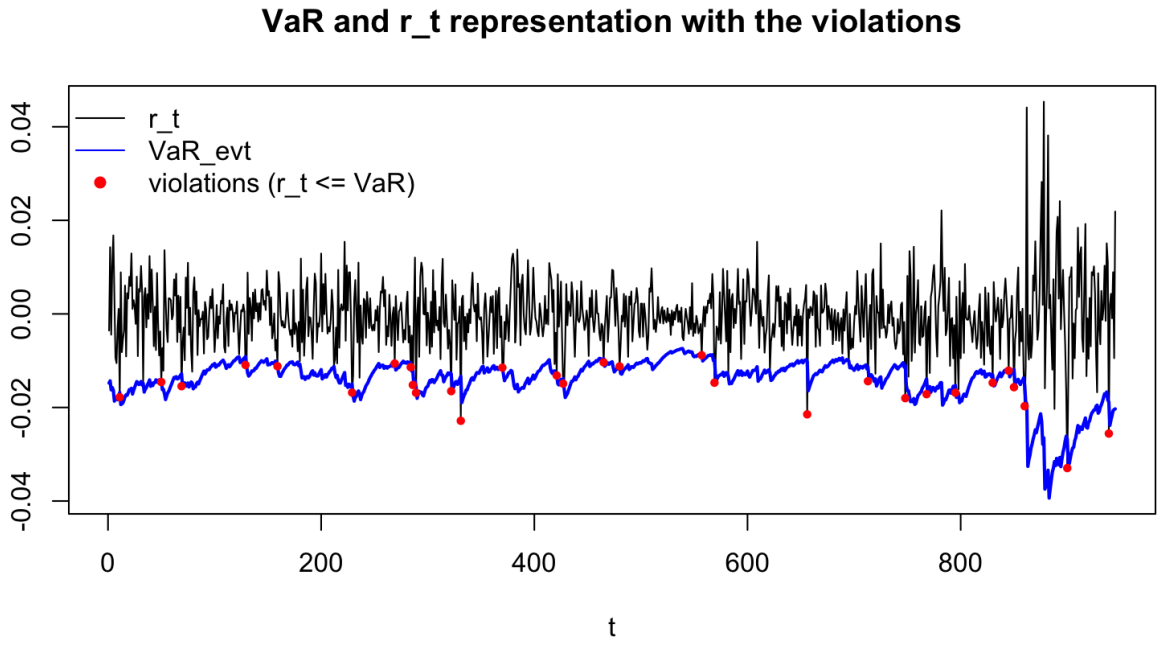


Figure 6: VaR_evt and r_t with violations (red dots).

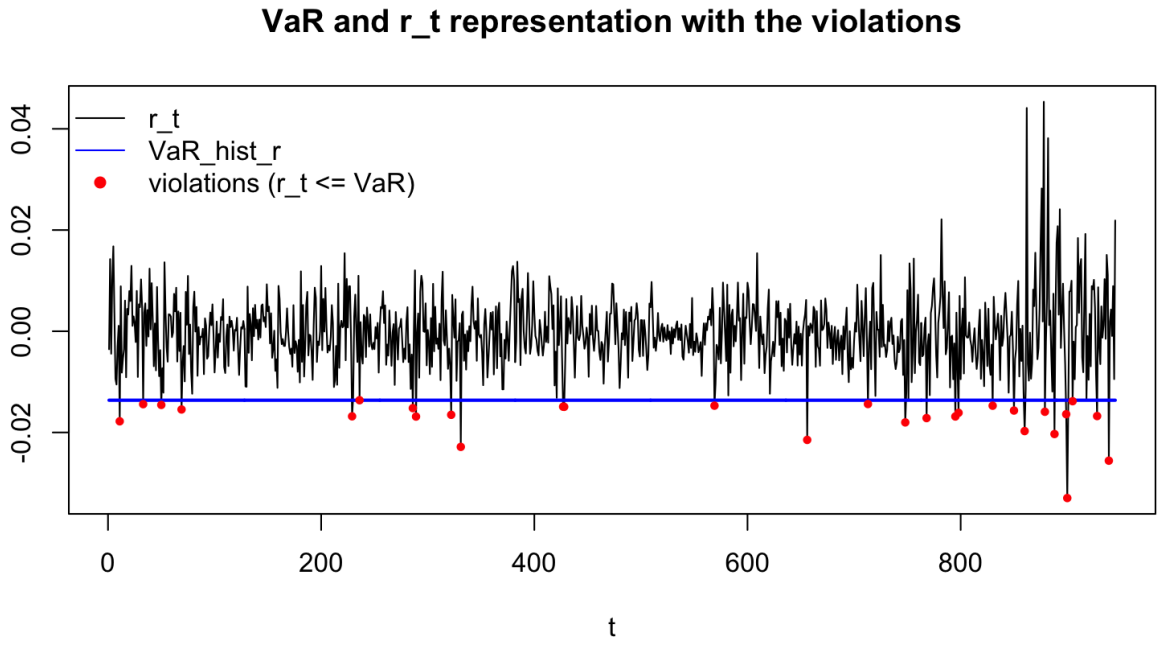


Figure 7: VaR_hist_r and r_t with violations (red dots).

Looking at the different graphs, we can see that the VaR evolves in a broadly similar way across all models. This is consistent with the fact that their mean values are close. Moreover, the number of violations seems to decrease when the VaR increases. We also observe that the `EVT` and the **historical** VaR lead to a similar number of violations. Likewise, the **Student** and **Gaussian** VaR produce approximately the same number of violations, but in this case the

number of violations is larger than for the `EVT` and historical VaR. This confirms our previous results.

2.4.2 Coverage Test

In this part we implement the **Christoffersen's (2004, Ch. 8) VaR backtesting** using the *hit sequence* (VaR violations) built from realized returns `r_t` and predicted VaR_t at level p (default $p = 0.03$). First, it checks that `r` and `VaR` have the same length and defines the violation indicator

$$I_t = \mathbf{1}\{r_t \leq \text{VaR}_t\},$$

so $I_t = 1$ when the realized return falls below the VaR forecast (a breach), and $I_t = 0$ otherwise. Over T observations, the code counts breaches

$$T_1 = \sum_{t=1}^{T} I_t$$

and non-breaches $T_0 = T - T_1$, and computes the empirical breach rate $\hat{\pi} = T_1/T$. The helper `term(count, prob)` returns `count` $\log($`prob`$)$ but safely outputs 0 when `count = 0`, avoiding numerical issues of the form $0\log(0)$ in log-likelihood calculations.

The **Unconditional Coverage test** (LR_{uc}) checks whether the observed breach frequency matches the nominal level p. It compares the Bernoulli likelihood under the null $\pi = p$,

$$L(p) = (1-p)^{T_0} p^{T_1},$$

to the likelihood under the unrestricted MLE $\pi = \hat{\pi}$,

$$L(\hat{\pi}) = (1-\hat{\pi})^{T_0} \hat{\pi}^{T_1},$$

and forms the likelihood-ratio statistic

$$LR_{uc} = -2\log\left(\frac{L(p)}{L(\hat{\pi})}\right) \;\sim\; \chi^2(1),$$

with `p_uc = 1 - pchisq(LRuc, 1)`.

The **Independence test** (LR_{ind}) checks whether violations cluster over time (i.e., whether I_t is independent). The code creates lagged/current hit vectors `I_lag` and `I_now` and counts the four Markov transitions

$$T_{ij} = \#\{t :\; I_{t-1} = i,\; I_t = j\}, \quad i, j \in \{0, 1\},$$

stored as `T00, T01, T10, T11`. It then estimates the transition probabilities by MLE:

$$\hat{\pi}_{01} = \frac{T_{01}}{T_{00} + T_{01}}, \qquad \hat{\pi}_{11} = \frac{T_{11}}{T_{10} + T_{11}},$$

(with safeguards when denominators are zero). Under the Markov alternative, the likelihood is

$$L(\hat{\Pi}) = (1 - \hat{\pi}_{01})^{T_{00}} \hat{\pi}_{01}^{T_{01}} (1 - \hat{\pi}_{11})^{T_{10}} \hat{\pi}_{11}^{T_{11}}.$$

Under the independence restriction, the hit sequence is i.i.d. Bernoulli with probability $\hat{\pi}$, which is what `lnL_ind` encodes. The LR statistic is

$$LR_{ind} = -2 \log\left(\frac{L(\text{independence})}{L(\hat{\Pi})} \right) \sim \chi^2(1),$$

with `p_ind = 1 - pchisq(LRind, 1)`.

Finally, the **Conditional Coverage test** (LR_{cc}) jointly tests correct coverage and independence by adding the two components:

$$LR_{cc} = LR_{uc} + LR_{ind} \sim \chi^2(2),$$

and computes `p_cc = 1 - pchisq(LRcc, 2)`. The returned list reports the sample size T, the empirical hit rate $\hat{\pi}$, the transition counts, and the three LR statistics with their p-values.

Table 33: Christoffersen backtesting results

model	hit_rate	LRuc	p_uc	LRind	p_ind	LRcc	p_cc
VaR_hist	0.0307	0.0153	0.9017	1.8386	0.1751	1.8538	0.3958
VaR_gauss	0.0444	5.9199	0.0150	0.0100	0.9205	5.9298	0.0516
VaR_student	0.0444	5.9199	0.0150	0.0100	0.9205	5.9298	0.0516
VaR_evt	0.0317	0.0972	0.7552	1.9698	0.1605	2.0669	0.3558
VaR_hist_r	0.0307	0.0153	0.9017	1.1066	0.2928	1.1219	0.5707

The `VaR_hist` model has a hit rate of 0.0307, which is very close to the nominal level $p = 0.03$. Moreover, we do not reject the Unconditional Coverage, Independence, or Conditional Coverage tests at the 5% significance level (i.e., `p_uc`, `p_ind`, and `p_cc` are all above 0.05). This suggests that the model does not underestimate risk and that violations do not appear to be clustered. Since we do not reject `p_cc`, the model passes the overall Christoffersen validation at the 5% level.

The `VaR_gauss` and `VaR_student` models yield very similar results, with a hit rate of 0.0444, which is above $p = 0.03$. They do not reject the Independence test at the 5% level (so violations do not seem to cluster), but they reject the Unconditional Coverage test since `p_uc` < 0.05. Therefore, these models fail to achieve correct coverage, indicating that the VaR forecasts are

not well calibrated at the 3% level.

The `VaR_evt` model has a hit rate of 0.0317, which is also very close to p. As with `VaR_hist`, we do not reject `p_uc`, `p_ind`, or `p_cc` at the 5% level, suggesting correct coverage, no evidence of clustering, and a successful validation.

Finally, the `VaR_hist_r` model has a hit rate close to p (about 0.0307). It passes the Conditional Coverage test at the 5% level, and it does not reject the Independence test, indicating no clustering. However, it rejects the Unconditional Coverage test (`p_uc` < 0.05), which points to a coverage mismatch and suggests that the VaR is not perfectly calibrated at the 3% level.

Overall, based on these backtesting results, the `VaR_hist` and `VaR_evt` approaches appear to be the most satisfactory choices for estimating the 3% VaR in our setting.

3 Exercice 3 : GMM estimation of a demand and supply model

Motivation and Interest of the question:

This exercise relies on the working paper *"Price Competition and Endogenous Product Choice in Networks: Evidence from the US Airline Industry"* by Christian Bontemps, Cristina Gualdani, and Kevin Remmy. Traditional merger analysis often assume that firms' product offerings remain unchanged after a merger. However, in several industries, and especially in network industries such as the airline industry, mergers create opportunities for network adjustments. The merged entity may expand or contract its route network, and competitors may react by entering or exiting markets. These changes can affect demand, costs, and consumer welfare. This paper addresses this issue by estimating a structural model in which airlines endogenously choose their networks before competing in prices. The model allows for spillovers and is applied to study the 2013 merger between American Airlines and US Airways.

3.1 Question 1,2,3 : Data Description

The database used in the empirical paper *"Price Competition and Endogenous Product Choice in Networks: Evidence from the U.S. Airline Industry"* is composed of a 10% random sample of all airline tickets sold in the United States during the third quarter of 2011. The sample is restricted to air travel between the 85 largest metropolitan statistical areas (MSAs) in the U.S. The dataset excludes tickets involving one-way trips, multiple carriers, asymmetric itineraries, and flights with more than one stop. As a result, a product is defined as an airline–itinerary combination between two cities with at most one stop and a market is defined as a non-directional city pair. Tickets with the same airline–itinerary combination but with different fares are treated as the same product.

Each observation corresponds to a differentiated product within a market. The variable `Market` uniquely identifies the city-pair market, while `Origin` and `Dest` label the two endpoint cities. For connecting itineraries, `C1` records the intermediate connecting airport, whereas it is typically missing for nonstop products. `Firm` identifies the operating carrier supplying the product. `Price` is the product-level ticket price and represents the key endogenous variable in the demand estimation. It is computed as a passenger-weighted average fare after trimming extreme values.

`Pax` measures the number of passengers purchasing a given product in a market. `MarketSize` captures the potential size of the market and is also used to compute the outside option share, which represents consumers who do not purchase any of the observed airline products. Market size is computed as the geometric mean of the populations of the two endpoint cities using data from the U.S. Census Bureau.

Itinerary quality is summarized by `Stops`, which equals zero for nonstop flights and one for one-stop itineraries, and by distance-related variables. `DirectDistance` measures the great-circle distance between the two endpoint cities, while `ExtraMiles` captures the additional routing distance implied by connecting itineraries relative to the direct route. Finally, `Fuel_Cost_per_seat` is a cost shifter affecting marginal cost and therefore pricing.

The sample includes the major carriers : Delta Airlines, United Airlines, American Airlines, US Airways, and Southwest Airlines. The first four carriers operate under a hub-and-spoke network structure, while Southwest Airlines follows a hybrid system based on focus cities rather than a hub-and-spoke model. All remaining airlines are classified as Low-Cost Carriers or Other carriers, which are considered fringe competitors.

3.2 Question 4

For this question, we estimate the demand model using 2SLS with homogeneous passengers and a nested logit specification. Before doing so, we create new explanatory variables.

We rely on a nested logit model, following the working paper, with two nests: one for the airline products (the inside alternatives) and one for the outside option (travelling by other means).

The Log of the ratio between the share of product j and the share of the outside option. It is computed as :

$$\ln s_{j0} = \log\left(\frac{\texttt{Pax}_j}{\texttt{MarketSize} - \sum_{k\in\mathcal{J}} \texttt{Pax}_k}\right).$$

The within-group market share measures the proportion of consumers choosing product j conditional on choosing an inside alternative. It is computed as :

$$\ln s_{j|g} = \log\left(\frac{\texttt{Pax}_j}{\sum_{k\in\mathcal{J}} \texttt{Pax}_k}\right).$$

3.2.1 Additional Variables

To obtain the results of the 2SLS regression we had to compute additional variables using the code posted on Moodle, to make it work we computed additional variables :

- `Number of departures from each city` : According to the empirical paper, we also include the maximum number of direct flights offered at the itinerary's endpoints by the

same carrier supplying itinerary j, denoted `OriginConn`, equivalent to the variable `Non Stop Origin` in the working paper.

- `Distance and square distance` : According to the working paper, we include `Distance` and `Distance`^{2} as observed as demand shifters. In the following code, we compute the distance by taking the product of `DirectDistance` (in miles) and `ExtraMiles` and then standardize this variable to reduce the collinearity. We assume that `ExtraMiles` acts as a multiplier of the direct distance for connecting itineraries. This allows us to approximate more accurately the total distance travelled by passengers, which is typically larger than the direct distance when stops imply detours.

- `Dummy for company` : To include in the model the majors carriers we create dummies for the companies : United Airlines (UA), Delta Airlines (DL), American Airlines (AA), US Airways (US), and Southwest Airlines (WN). Finaly we also create dummy's for the LCC firm and the other firm. To do so we use the definition of the article saying that LCC firm are : Frontier Airlines, Alaska Airlines, Spirit Airlines, JetBlue Airlines, Virgin Airlines, Sun County Airlines, and Allegiant Air. For the other company : AirTran Airways and USA3000 Airlines.

- `City list` : We will also create this new variable city dummies based on orign and destination airports. In the working paper, they have added carrier and city as fixed effects.

3.2.2 Instruments

- `Dummy for monopoly and Number of competing products` : The dataset already contains `Fuel_Cost_per_seat`. In the following code, we create (i) a monopoly dummy, (ii) the number of competing products, and (iii) the number of competitors of the same type (LCC and FSC). By definition, a market is a monopoly when only one firm is present in that market. Therefore, we construct the monopoly dummy by counting the number of distinct firms operating in each market (where a market is defined as a non-directional city pair). The number of competing products is defined as the total number of products offered in a given market excluding the product itself.

- `Number of competitors of the same type (LCC vs. FSC)` : We construct the number of same-type competitors by counting, within each market, how many airlines belong to the same carrier type (LCC vs. FSC) and subtracting one to exclude the firm itself. We include in the FSC group the four main legacy carriers and the `Others` group. The LCC group consists of carriers classified as Low-Cost Carriers. We exclude Southwest Airlines (`WN`) because, in the working paper, `WN` is characterized as a distinct hybrid carrier: it is neither one of the four main legacy carriers nor part of the `Others` or `LCC` groups. Therefore, to be consistent with the paper, we define three categories: `LCC`, `FSC`, and `WN`.

Additionally, we assume the following carrier codes:

- B6=JetBlue,
- NK=Spirit Airlines,
- F9=Frontier Airlines,
- VX=Virgin America,
- SY=Sun Country Airlines,
- G4=Allegiant Air,
- U5=USA3000 Airlines,
- AS=Alaska Airlines,
- FL=AirTran Airways,
- UA=United Airlines,
- DL=Delta Airlines,
- AA=American Airlines,
- US=US Airways,
- WN=Southwest Airlines.

3.2.3 Results

The following 4 tables display the results of the 4 models with different combinaisons of the three instruments:

Table 34: Model 1 results

stat	Value
instruments	Fuel_Cost_per_seat + Monopoly + SameTypeCompetitors
mean_own_price_elas	28.1004
mean_lerner	0.0054
price_coef	-0.01208
sigma	1.1262
J_stat	11.346
J_pval	8e-04

Table 35: Model 2 results

stat	Value
instruments	Fuel_Cost_per_seat + Monopoly + CompetingProducts
mean_own_price_elas	-4.7376
mean_lerner	0.2412
price_coef	-0.0063
sigma	0.5141
J_stat	221.4434
J_pval	0

Table 36: Model 3 results

stat	Value
instruments	Fuel_Cost_per_seat + CompetingProducts + SameTypeCompetitors
mean_own_price_elas	-3.8471
mean_lerner	0.2962
price_coef	-0.0052
sigma	0.5033
J_stat	256.6369
J_pval	0

Table 37: Model 4 results

stat	Value
instruments	CompetingProducts + Monopoly + SameTypeCompetitors
mean_own_price_elas	-2.084
mean_lerner	0.5248
price_coef	-0.0039
sigma	0.2678
J_stat	1.6835
J_pval	0.1945

Table 38: Median own-price elasticities by model

Model	Median_own_price_elasticity
Model 1	29.7932
Model 2	-4.6096
Model 3	-3.7417
Model 4	-2.0294

After having computed the mean/median own-price elasticity, estimates of the price coefficient and sigma, the mean Lerner Index, the J test statistic on all the possible combinations of three instruments, including Fuel cost per seat, dummy for monopoly, number of competing products, number of competitors of the same type. We have to select the model that gives the most coherent estimates.

Instrument selection. In order to select the best set of instruments, we compare our estimates with the benchmark results reported in the reference paper. The paper reports a mean own-price elasticity of -3.782, a price coefficient of -0.587, and a mean Lerner index ranging between 0.27 and 0.31.

Model 1 yields a positive own-price elasticity of 28, which is economically implausible because it would imply that demand increases with price. We therefore exclude Model 1 from the analysis. Turning to the price coefficient, we observe that its estimated value is far from the benchmark -0.587 in all remaining models; consequently, this criterion does not allow us to discriminate reliably across specifications. This can be explained by the fact that we are currently working with a subset of the data used in the working paper, hence it can explained the divergence in the results. Since the price coefficients in Models 2–4 are all negative, we cannot eliminate any of them based on the sign either.

By contrast, the mean own-price elasticity in Model 3 is -3.847, which is very close to the benchmark value of -3.782. Models 2 and 4 produce mean elasticities that are approximately one unit away from the benchmark. Moreover, the mean Lerner index in Model 3 is 0.29, which is also close to the benchmark range [0.27, 0.31].

Finally, we consider the Sargan overidentification test. When the p-value is below 0.05, we reject the null hypothesis that all instruments are jointly valid. In our results, only Model 4 has a p-value above 0.05, implying that it is the only specification for which we do not reject instrument validity. Although Model 3 appears to match the benchmark elasticities and markups more closely, it rejects the null and therefore cannot be selected as the most credible instrument subset. We consequently retain Model 4 as the preferred specification. While its estimates are less close to the benchmark values, they remain economically plausible and not

inconsistent with the model.

3.3 Question 5

For this question we have to first extract the estimated marginal costs from the model 4 (as chosen previously in Question 4) derived previously and regress it on the "usual characteristics".

According to the working paper, the marginal costs are derived as the difference between the prices and the markups. We rely on the technique used in the paper and they expressed the marginal costs as a function of observed and unobserved costs shifters :

$$MC_{j,m} = W'_{j,m}\theta_s + \omega_{j,m},$$

where $W_{j,m}$ is a vector of marginal cost shifters observed by the researcher and $\omega_{j,m}$ captures marginal cost shifters that are unobserved by the researcher. The paper includes in $W_{j,m}$ various characteristics such as the number of stops, the distance flown, and whether the itinerary is short-haul or long-haul, along with carrier fixed effects and the number of cities that are reachable from the endpoints and intermediate stops of itinerary j with the same carrier offering itinerary j , called `Connections`.

We already have in mydata the variables `Stop` and `Distance`, we have to create a dummy variable for short and long-haul itinerary and the variable `Connections`.

First, to introduce the variables `Short` and `Long` in the equation, we follow the methodology of the working paper and classify flights according to their length. Specifically, we define short-haul flights as routes covering up to 1,500 miles, and long-haul flights as routes covering more than 1,500 miles. To define the short- and long-haul categories, we base the classification on the variable `DirectDistance` rather than `Distance`, in order to remain fully consistent with the description in the paper, where distance is defined as the geographical distance between the origin and the destination rather than the flown distance.

Secondly, concerning the variable `Connections`, we follow the working paper for the construction of this variable. For each airline and city, we first compute the number of distinct destinations reachable either directly or with one stop. We defined the variable `Connections` as the average number of reachable destinations across the itinerary (Origin, Destination and Intermediate stop).

Finally, we will also include the dummies for companies for the fixed effects in the equation.

After constructing all the required variables, we estimate the marginal cost regression using the marginal cost estimates obtained in the previous question.

3.3.1 Results

NB. The units of measurement used in our estimates differ from those in the working paper and are therefore not comparable. In the working paper, marginal costs are expressed in hundreds of U.S. dollars, the variable `Connections` is measured in hundreds of cities, and `Distance` is expressed in thousands of miles.

Looking at the Table 39 it can be seen that the median Marginal Cost is equal to 206,5 and that the average one is equal to 215,7. Also it can be seen that some firm have a negative marginal cost this can correspond to companies with not fully booked flight, so this is not uncoherent.

Now, looking at the regression coefficients (Table 40), we can see that some variables increase the marginal cost more than others.

For example, a flight with an LCC carrier decreases the marginal cost by approximately 103 USD ceteris paribus. However, this is not the case for all carriers, as exemple a flight with WN decrease the marginal cost by 31 USD, on the other hand having a flight with US Airways or Delta Airlines increase the marginal cost by 17 USD for US Airways and 8 USD for Delta Airlines. Finaly flight of "Other" careers decrease the marginal cost by 48 USD.

Distance also plays a role in the magnitude of the marginal cost. An increase of one mile flown in the distance variable increases the marginal cost by 6 cents for short-distance flights and 8 cents for long-distance flights, ceteris paribus. At the unitary level, this is small, but we must keep in mind that flights cover thousands of miles, so an increase in distance is often hundreds of miles. This can be explained by increased fuel consumption: the greater the distance covered by the flight, the more fuel it will burn, which increases the overall cost of the flight as well as the marginal cost.

The 'indirect' variables has a significant effect if the flight is considered short, but not if it is considered long. In the case of a short flight, having an indirect flight increases the marginal cost by approximately 10 USD, ceteris paribus. This effect may be explained by the fact that a stop is necessary on a long flight due to the fuel reserve. On the other hand, for a short flight, the fuel reserve does not necessarily need to be filled, so it may induce unnecessary costs for the carrier, as the flight would need to land and take off twice and would not be able to take the fastest route to the final destination. Additionally, it cannot be excluded that landing at an airport incurs a cost for the carrier, which would need to be paid twice.

Finally, the 'connections' variables decrease the marginal cost for both long and short flights. For short flights, this decrease is 0.5 USD for each additional destination covered by a carrier, and 0.9 USD for long flights, ceteris paribus. This reduction may be due to the fact that the

more cities that are covered, the more likely it is that the planes will be full after a connection at another airport.

Table 39: Summary statistics of estimated marginal costs

Statistic	Value
Min.	-114.2
1st Qu.	137.3
Median	206.5
Mean	215.7
3rd Qu.	284.6
Max.	705.6

Table 40: Estimated Marginal Cost

	Dependent variable:
	mc_hat
Intercept_Short	163.719***
	(8.039)
Intercept_Long	171.904***
	(13.130)
Indirect_Short	10.477***
	(2.614)
Indirect_Long	8.534
	(5.231)
Distance_Short	0.058***
	(0.003)
Distance_Long	0.080***
	(0.005)
Connections_Short	−0.543***
	(0.143)
Connections_Long	−0.928***
	(0.160)
DL	8.064*
	(4.160)
UA	3.451
	(3.091)
US	17.307***
	(3.418)
WN	−31.471***
	(3.459)
LCC	−102.919***
	(4.261)
others	−48.545***
	(10.478)
Observations	10,915
R^2	0.861
Adjusted R^2	0.861
F Statistic	4,817.472*** (df = 14; 10901)
Note:	*p<0.1; **p<0.05; ***p<0.01

3.4 Question 6

For this question, we had to re-estimate the model using GMM. We have to compare the optimal GMM with the basic model using $W = I_d$.

We have used and adapted the code given on Moodle, we estimate the demand and supply parameters jointly by GMM and we report in the following table both the optimal GMM and the basic model with $W = I_d$.

3.4.1 Results

Table 41: Estimated parameters

Parameter	OLS/2SLS	GMM_WI	GMM_opt	p_OLS_2SLS	p_WI	p_opt
D: Intercept	-4.435***	-4.435***	-4.433***	0.000	0.000	0.000
	(0.514)	(0.000)	(0.040)			
D: Price	-0.004***	-0.004***	-0.004***	0.000	0.000	0.000
	(0.001)	(0.000)	(0.000)			
D: lnsjg	0.268***	0.268***	0.268***	0.000	0.000	0.000
	(0.050)	(0.000)	(0.006)			
D: Stops	-1.937***	-1.937***	-1.939***	0.000	0.000	0.000
	(0.114)	(0.000)	(0.028)			
D: OriginConn	0.006***	0.006***	0.006***	0.000	0.000	0.000
	(0.001)	(0.000)	(0.001)			
D: Distance	-0.008	-0.009***	-0.008	0.879	0.000	0.453
	(0.054)	(0.000)	(0.011)			
D: Distance2	-0.068***	-0.068***	-0.068***	0.000	0.000	0.000
	(0.011)	(0.000)	(0.008)			
D: DL	-0.062*	-0.062***	-0.062***	0.063	0.000	0.004
	(0.034)	(0.000)	(0.021)			
D: UA	-0.322***	-0.322***	-0.322***	0.000	0.000	0.000
	(0.040)	(0.000)	(0.023)			
D: US	0.094*	0.094***	0.094***	0.075	0.000	0.000
	(0.053)	(0.000)	(0.024)			
D: WN	-0.242***	-0.242***	-0.242***	0.000	0.000	0.000
	(0.040)	(0.000)	(0.024)			
D: LCC	-0.134*	-0.134***	-0.134***	0.060	0.000	0.000
	(0.071)	(0.000)	(0.026)			
D: others	-0.436***	-0.436***	-0.436***	0.000	0.000	0.000
	(0.111)	(0.000)	(0.015)			

S: Intercept_Short	163.719***	163.719***	163.653***	0.000	0.000	0.000
	(8.039)	(0.000)	(0.000)			
S: Intercept_Long	171.904***	171.904***	171.917***	0.000	0.000	0.000
	(13.130)	(0.000)	(0.000)			
S: Indirect_Short	10.477***	10.477***	10.473***	0.000	0.000	0.000
	(2.614)	(0.000)	(0.000)			
S: Indirect_Long	8.534	8.534***	8.563***	0.103	0.000	0.000
	(5.231)	(0.000)	(0.000)			
S: Distance_Short	0.058***	0.058***	0.058***	0.000	0.000	0.000
	(0.003)	(0.000)	(0.003)			
S: Distance_Long	0.080***	0.080***	0.080***	0.000	0.000	0.000
	(0.005)	(0.001)	(0.003)			
S: Connections_Short	-0.543***	-0.543***	-0.543***	0.000	0.000	0.000
	(0.143)	(0.000)	(0.059)			
S: Connections_Long	-0.928***	-0.928***	-0.928***	0.000	0.000	0.000
	(0.160)	(0.000)	(0.036)			
S: DL	8.064*	8.064***	8.070***	0.053	0.000	0.000
	(4.160)	(0.000)	(0.001)			
S: UA	3.451	3.451***	3.445***	0.264	0.000	0.000
	(3.091)	(0.000)	(0.000)			
S: US	17.307***	17.307***	17.286***	0.000	0.000	0.000
	(3.418)	(0.000)	(0.000)			
S: WN	-31.471***	-31.471***	-31.534***	0.000	0.000	0.000
	(3.459)	(0.000)	(0.000)			
S: LCC	-102.919***	-102.919***	-102.930***	0.000	0.000	0.000
	(4.261)	(0.000)	(0.001)			
S: others	-48.545***	-48.545***	-48.523***	0.000	0.000	0.000
	(10.478)	(0.000)	(0.000)			

We can see that the coefficients of all three models are very close, almost identical, for both demand and supply. These results could be due to the fact that the classic GMM model is already very close to the optimal one, or because we have only just identified m-k, which is equal to 0 or very close to it. Finally, this could also be due to the choice of the instrumments from our model 4 (of the question 4). Even though the instruments we chose were the only ones to reject H0 (at least one of the instruments is not valid).

3.4.2 Demand coefficients

Looking at the coefficients for our three models, it is clear that the majority of them are significant at the 99% level. Only one is not significant: distance for both the OLS/2SLS and the

optimal GMM models.

The first thing to consider is the price. Here, an increase of one dollar in the flight price decreases demand by almost half a point. This is consistent with the idea that an increase in price should not increase demand.

Another variable that has a significant impact on demand is the number of stops. A flight with a stop decreases demand by almost two points, which seems coherent since people taking a flight will want to go directly to their final destination.

Finally, when looking at the different carriers, it is clear that the effects differ across carriers. For example, LCCs and other carriers decrease demand for flights, whereas major carriers (except US airlines) increase demand. These results may be due to the carriers' onboard standards or the destinations they cover.

3.4.3 Supply coefficients

As for the demand, the coefficients on the supply side are almost all significant at the 99% level of confidence. Only 'indirect_long' and UA are not significant for the OLS/2SLS model.

As with question 5, it can be seen that increasing the distance by one mile increases the marginal cost, ceteris paribus. This is coherent with the previous argument, since increasing the distance increases fuel consumption for both long- and short-distance flights.

Additionally, having an indirect flight increases the marginal cost by approximately 10 USD, ceteris paribus, for all models. This is also consistent with our findings for question 5. As for question 5, the connection variable decreases the marginal cost.

Finally, for the carrier variables, we retrieve the same results as before for all the models: a flight with an LCC carrier decreases the marginal cost by approximately 103 USD, ceteris paribus. However, this is not the case for all carriers: for example, a flight with WN decreases the marginal cost by 31 USD. On the other hand, a flight with US Airways or Delta Airlines increases the marginal cost by 17 USD and 8 USD respectively. Finally, flights with "other" carriers decrease the marginal cost by 48 USD.